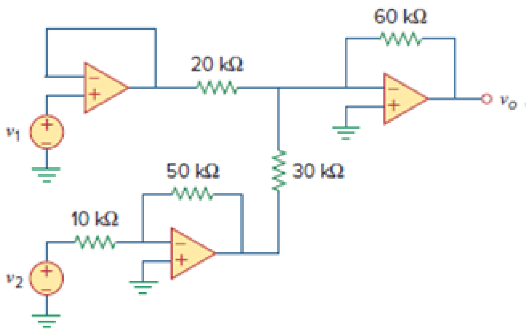
Chapter 5, Problem 10PP

Problem

If $v1$ = 5 V and $v2$ = 5 V, find vo in the op amp circuit of Fig. 5.33.

Figure. 5.33:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 5.33 in the textbook.

The circuit with node notation is shown in Figure 1.

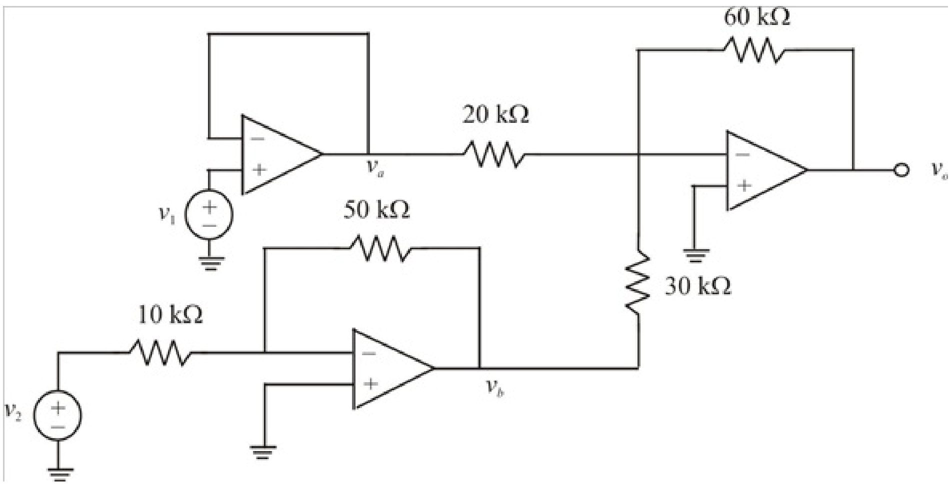


Figure 1

Step 2 of 3

The voltages at inverting and non-inverting op-amps are same for ideal op-amps.

The first op-amp is a voltage follower. The voltage at node *a* is,

$$\begin{aligned} v_a &= v_1 \\ &= 5 \text{ V} \end{aligned}$$

The second op amp is an inverting amplifier.

The output voltage of the inverting amplifier is,

$$v_b = -\frac{R_f}{R_1} v_i$$

Substitute $50 \text{ k}\Omega$ for R_f, $10 \text{ k}\Omega$ for R_1, and 5 V for v_2 in the above equation.

$$\begin{aligned} v_b &= -\left(\frac{50 \text{ k}\Omega}{10 \text{ k}\Omega}\right)(5 \text{ V}) \\ &= -(5)(5 \text{ V}) \\ &= -25 \text{ V} \end{aligned}$$

Step 3 of 3

The third op-amp is a summing amplifier.

Calculate the output voltage v_o of the third op amp.

$$v_o = \left(\frac{-R_f}{R_1}\right) v_a + \left(\frac{-R_f}{R_2}\right) v_b$$

Substitute $60 \text{ k}\Omega$ for R_f, $20 \text{ k}\Omega$ for R_1, $30 \text{ k}\Omega$ for R_2, 5 V for v_a, and -25 V for v_b in the above equation.

$$\begin{aligned} v_o &= \left(\frac{-60 \text{ k}\Omega}{20 \text{ k}\Omega}\right)(5 \text{ V}) + \left(\frac{-60 \text{ k}\Omega}{30 \text{ k}\Omega}\right)(-25 \text{ V}) \\ &= (-3)(5 \text{ V}) + (-2)(-25 \text{ V}) \\ &= -15 \text{ V} + 50 \text{ V} \\ &= 35 \text{ V} \end{aligned}$$

Therefore, the output voltage, v_o is $\boxed{35 \text{ V}}$.